

ZI-E Lite Aluminum v0.1

Software-first stationary prototype - fixed head - no 3D printing

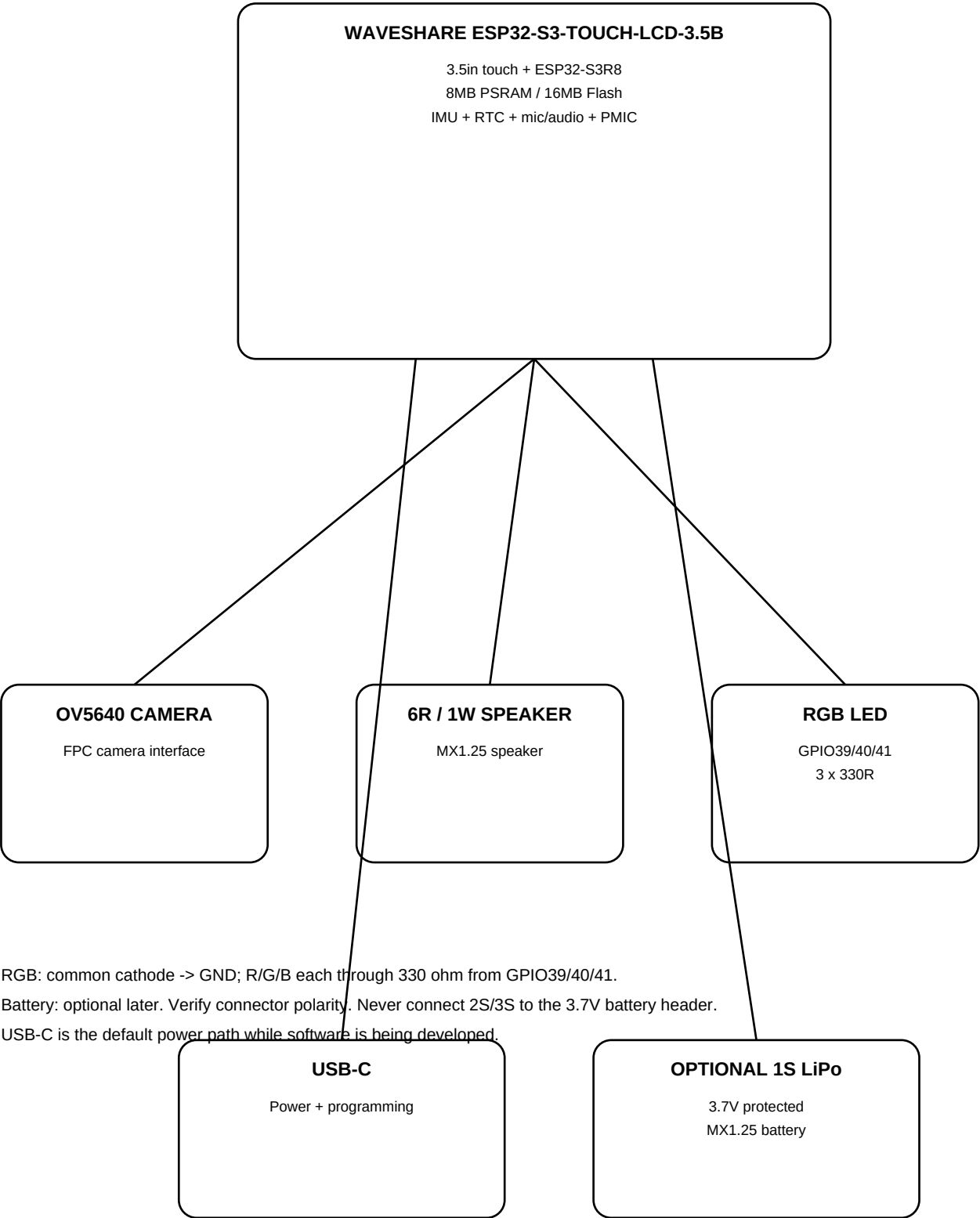
Build objective: finish the AI/face/vision/voice software in a real ZI-E shell before buying the expensive wheel, arm and motorized-head mechanism.

Nominal size: body 126 mm high, 158->146 mm wide, 108->98 mm deep. Head 128->120 x 90 x 40 mm.

Preferred electronics: Waveshare ESP32-S3-Touch-LCD-3.5B + OV5640. One integrated board replaces the separate display, ESP32 dev board and camera module.

Critical aluminum-body fix: reserve a non-metallic RF window in the head. Do not trap the ESP32 2.4 GHz antenna in a closed metal box.

Wiring block diagram



Assembly order

1. Print the cut-template PDF at 100% and verify the 50 mm calibration line.
2. Cut 1.5 mm 5052-H32 shell panels and the 2.0 mm stationary base. Keep rear body/head panels removable.
3. Dry-fit and clamp. Check symmetry and diagonal measurements. Do not install electronics.
4. Tack opposite corners, alternating sides. Recheck screen opening and rear cover fit.
5. Finish short welds, cool, deburr and remove all metal chips.
6. Fit the non-metal RF window, grommets, rubber feet and insulating PCB carrier.
7. Install the Waveshare board on M2 standoffs, camera FPC, speaker and RGB LED.
8. Power by USB-C only. Run board factory diagnostics before custom firmware.
9. Complete face/touch/camera/audio/Wi-Fi/AI software.
10. Only then buy/install the optional 1S battery. Mechanical Full-version parts remain a later branch.